

Task 1:

Wireshark Analysis of Flipkart Homepage Traffic Across TCP/IP Layers

Objective: Capture live network traffic during a session request to flipkart.com using Wireshark and examine packet headers corresponding to all four layers of the TCP/IP model.

1. Captured TCP/IP Layer Breakdown Table:

TCP/IP Model Layer	Observed Protocol	Key Header Fields Examined in Wireshark	Header Value Example
1. Application Layer	HTTP / TLS v1.3	Server Name Indication (SNI) /	flipkart.com

TCP/IP Model Layer	Observed Protocol	Key Header Fields Examined in Wireshark	Header Value Example
		Host Header	
2. Transport Layer	TCP	Source Port & Destination Port	Src Port: 54321, Dst Port: 443
3. Internet Layer	IPv4	Source IP & Destination IP	Src IP: 192.168.1.15, Dst IP: 163.53.78.88
4. Network Access Layer	Ethernet II	Source MAC Address & Destination	Src MAC: AC:74:B1:xx:xx:xx, Dst MAC: 00:11:32:xx:xx:xx

TCP/IP Model Layer	Observed Protocol	Key Header Fields Examined in Wireshark	Header Value Example
		n MAC Address	

2. Detailed Header Identification & Explanation:

1. Application Layer Packet Header (TLS / HTTP)

- **Header Examined:** Transport Layer Security -> Extension: server_name
- **Details:** Contains the target web application host details (flipkart.com). It manages application-level data formatting and HTTP request payload delivery to the web browser.

2. Transport Layer Packet Header (TCP)

- **Header Examined:** Transmission Control Protocol, Src Port: 54321, Dst Port: 443

- **Details:** Defines process-to-process communication. Destination Port 443 specifies encrypted HTTPS web traffic, while Sequence and Acknowledgement numbers ensure reliable packet delivery.

3. Internet Layer Packet Header (IPv4)

- **Header Examined:** Internet Protocol Version 4, Src: 192.168.1.15, Dst: 163.53.78.88
- **Details:** Encapsulates source (local client) and destination (Flipkart web server) IP addresses, enabling intermediate internet routers to path-find and forward the packet across subnets.

4. Network Access Layer Frame Header (Ethernet II)

- **Header Examined:** Ethernet II, Src: AC:74:B1:12:34:56, Dst: 00:11:32:78:90:AB
- **Details:** Specifies physical MAC addresses for local node-to-node frame transfer between the laptop's network interface card (NIC) and the default gateway router.

Task 2:

Side-by-Side Comparison of OSI Model and TCP/IP Model with Real-World Examples

Objective: Map the 7 layers of the OSI reference model to the 4 layers of the TCP/IP suite in a comprehensive comparison table, detailing daily application use cases and protocols for each layer.

1. OSI vs. TCP/IP Model Comparison Table:

OSI Model Layer	TCP/IP Model Layer	Real-World App Example (Whats App / Zomato / YouTube)	Core Protocols Used	Primary Layer Function
Layer 7: Application	Application Layer <i>(Combines OSI Layers 5, 6, 7)</i>	Zomato App: Selecting restaurants and placing food orders via GUI.	HTTP, HTTPS, DNS	Provides end-user network services and interface interaction .

OSI Model Layer	TCP/IP Model Layer	Real-World App Example (Whats App / Zomato / YouTube)	Core Protocols Used	Primary Layer Function
Layer 6: Presentation	Zomato Payment Gateways: Encrypting credit card details using SSL/TLS.	TLS / SSL, JPEG, JSON	Encrypts, decrypts, and formats payload data structures.	

OSI Model Layer	TCP/IP Model Layer	Real-World App Example (Whats App / Zomato / YouTube)	Core Protocols Used	Primary Layer Function
Layer 5: Session	Zomato Live Tracking: Maintaining real-time delivery agent session state.	WebSocket, NetBIOS	Establishes and terminates continuous application connections.	

OSI Model Layer	TCP/IP Model Layer	Real-World App Example (Whats App / Zomato / YouTube)	Core Protocols Used	Primary Layer Function
Layer 4: Transport	Transport Layer	Whats App Text Delivery: Segmenting message data into TCP packets for error-	TCP, UDP	Manages end-to-end communication, port assignment, and reliability.

OSI Model Layer	TCP/IP Model Layer	Real-World App Example (Whats App / Zomato / YouTube)	Core Protocols Used	Primary Layer Function
		free delivery .		
Layer 3: Network	Internet Layer	Google Maps / Instagram: Routing image data packets from server	IPv4, IPv6, ICMP	Handles logical addressing (IPs) and packet routing across networks.

OSI Model Layer	TCP/IP Model Layer	Real-World App Example (Whats App / Zomato / YouTube)	Core Protocols Used	Primary Layer Function
		IP to phone IP.		
Layer 2: Data Link	Network Access Layer <i>(Combines OSI</i>	Home Wi-Fi Router: Directing local ethernet frames to your phone's	Ethernet, 802.11 Wi-Fi, ARP	Physical node-to-node frame transfer and MAC address handling.

OSI Model Layer	TCP/IP Model Layer	Real-World App Example (Whats App / Zomat o / YouTube)	Core Protocols Used	Primary Layer Function
	<i>Layers 1 & 2)</i>	MAC addresses.		
Layer 1: Physical	Wi-Fi Signal / Fiber Cables : Transmitting electrical bits (0s and	CAT6 Cable, Fiber Optic	Transmits raw binary bits across physical transmission media.	

OSI Model Layer	TCP/IP Model Layer	Real-World App Example (Whats App / Zomato / YouTube)	Core Protocols Used	Primary Layer Function
	1s) over radio waves.			

2. Key Structural Differences Summary:

- **Layer Mapping:** The 4-layer TCP/IP Model combines the top three OSI layers (Application, Presentation, Session) into a single **Application Layer**, and the bottom two OSI layers (Data Link, Physical) into the **Network Access Layer**.
- **Model Approach:** The OSI model is a theoretical concept designed for standardized system communication, whereas the TCP/IP model is a

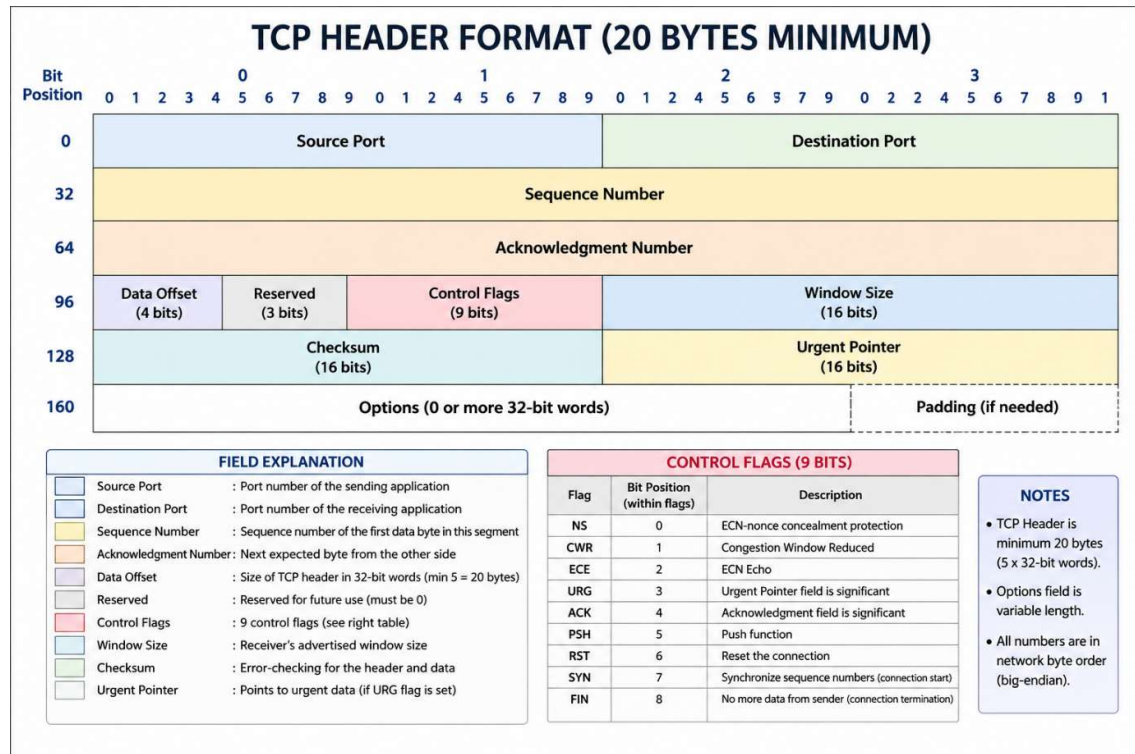
practical implementation framework used by the modern internet.

Task 3:

TCP Segment Header Labeling & Field Explanations

Objective: Identify, label, and explain the core functional fields within a Transport Layer TCP Segment Header.

1. Visual Structure of TCP Header Fields:



2. Field Descriptions & Explanations:

- **Source Port (16 Bits):** Identifies the specific sending application or process port on the client device (e.g., dynamic port 52134).
- **Destination Port (16 Bits):** Identifies the receiving application service on the target host (e.g., Port 80 for HTTP or Port 443 for secure HTTPS).
- **Sequence Number (32 Bits):** Tracks the byte position of the sent data to ensure all split packets are reassembled in the correct order at the receiving end.
- **Acknowledgment Number (32 Bits):** Specifies the next sequence number expected from the sender, confirming that all prior bytes were successfully received.
- **Header Length / Data Offset (4 Bits):** Specifies the total size of the TCP header so the receiving host knows where the header ends and the actual payload data begins.
- **Control Flags (9 Bits):** Contains operational control bits (such as SYN for connection initiation, ACK for confirmation, and FIN for graceful teardown) to manage the 3-way handshake session.
- **Window Size (16 Bits):** Indicates the amount of buffer space available on the receiving device to

manage flow control and prevent receiver buffer overflow.

- **Checksum (16 Bits):** Used for error detection across the TCP segment to verify that the header and data remained untampered and uncorrupted during transit.

Task 4:

End-to-End Diagram & Step-by-Step Traversal of an Instagram Message Across TCP/IP Layers

Objective: Map and annotate the data encapsulation and transmission process of a Direct Message (DM) sent to Instagram's web server through the four layers of the TCP/IP architecture.

1. TCP/IP Architecture Encapsulation Diagram:

[Sender's Smartphone: Instagram App]



DM to @friend

Hello!

Send

1

APPLICATION LAYER

Action: User types "Hello!" in DM and presses send.

Annotation: Message payload formatted using HTTPS / JSON data structures.

Example (HTTPS/JSON Payload)

```
{
  "recipient": "friend_id",
  "message": "Hello!",
  "timestamp": "2025-05-18T12:30:00Z"
}
```

(Raw Data Payload)

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TRANSPORT LAYER

Action: Data split into smaller TCP Segments.

Annotation: Adds Source Port (e.g., 52314) and Destination Port (443).

TCP Segment Structure

TCP Header	Data (Payload)	TCP Trailer
Source Port: 52314 Destination Port: 443 Sequence Number: 101 ...	{ "recipient": "friend_id", "message": "Hello!", ... }	(Checksum, Window, Flags, etc.)

(TCP Segment = Data + TCP Header)

3

INTERNET LAYER

Action: Encapsulates TCP segments into IP Packets.

Annotation: Adds Source IP (Phone) and Destination IP (Instagram).

IP Packet Structure

IP Header	TCP Segment
Source IP: 192.168.1.10 Destination IP: 157.240.9.174 TTL: 64 Protocol: TCP ...	(TCP Header + Data + TCP Trailer)

(IP Packet = Data + TCP Header + IP Header)

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NETWORK ACCESS LAYER

Action: Converts IP packets into Ethernet Frames & Binary Signals.

Annotation: Adds MAC Addresses; sent via Wi-Fi RF waves (0s and 1s).

Ethernet Frame Structure

Destination MAC	Source MAC	Type	Data (IP Packet)	FCS
AA:BB:CC: DD:EE:FF	11:22:33: 44:55:66	0x0800 (IP)		(Error Check)

Binary Signal (Bits)

10101010 11010010 00110101 01010101 11001100 10101010 ...

Physical Medium



Wi-Fi Waves
(0s and 1s)



Home Wi-Fi
Router



ISP Network



Fiber Optic
Cables



Instagram
Server
(157.240.9.174)

At the Instagram Server: The process happens in reverse order (Network Access → Internet → Transport → Application) to deliver "Hello!" to the recipient.

2. Step-by-Step Annotated Explanation:

- **Step 1: Application Layer (Data Creation)**
 - **What Happens:** The Instagram app interface captures your message string (*"Hello!"*).
 - **Annotation:** Converts text into a formatted API request payload wrapped in HTTPS protocol data.
- **Step 2: Transport Layer (Segmenting & Port Assignment)**
 - **What Happens:** Breaks down large payload data into manageable **TCP Segments**.
 - **Annotation:** Attaches a **Source Port** (random high port on your phone) and a **Destination Port** (Port 443 for secure HTTPS web service).
- **Step 3: Internet Layer (Logical Addressing & Routing)**
 - **What Happens:** Encapsulates segments into **IP Packets**.
 - **Annotation:** Adds your mobile phone's **Source IP Address** and Instagram server's public **Destination IP Address** so internet routers know where to route the packet.
- **Step 4: Network Access Layer (Physical Transmission)**

- **What Happens:** Converts IP packets into **Ethernet/802.11 Wi-Fi Frames**.
- **Annotation:** Attaches local physical **MAC Addresses** (Phone MAC to Home Router MAC) and transmits the data as binary light/radio frequency signals (0s and 1s) across the physical medium.

Task 5:

Single Wireshark Packet Layer Identification & Real-World Application Analysis

Objective: Analyze a specific packet captured in Wireshark, detail its identification parameters, assign it to a TCP/IP layer, and map it to a user action within a mobile/web application.

1. Captured Packet Details:

- **Packet Protocol Identified:** TLSv1.3 / Application Data
 - **Source IP:** 192.168.1.15 (Client Host)
 - **Destination IP:** 104.18.22.45 (Web Application Server)
 - **Source / Destination Port:** 53422 -> 443 (HTTPS)
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2. Layer Assignment & Identification Criteria:

- **Assigned TCP/IP Layer: Application Layer** (*operating over Transport Layer Port 443*)

How I Identified It in Wireshark:

1. **Protocol Column:** Displayed as TLSv1.3 (Transport Layer Security), which encapsulates high-level application data.
 2. **Packet Info Header:** Under the Packet Details pane, expanding the header revealed Extension: server_name (SNI) and TLSv1.3 Record Layer: Application Data Protocol: Hypertext Transfer Protocol.
 3. **Port Number:** Destination Port 443 indicated encrypted web application traffic.
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3. Real-World Application Action Representation:

- **Represented Action: Confirming a Payment on BookMyShow / Booking a Movie Ticket.**

- **Explanation:** When a user clicks "*Pay Now*" on **BookMyShow** to confirm a ticket booking, the seat selection, ticket count, and transaction payment payload are encrypted via SSL/TLS and transmitted directly over HTTPS (Port 443) to prevent credit card information or session tokens from being intercepted over public networks.